

PHYS 230 - Electricity and Magnetism

Lecture 17

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

4 Capacitance and Dielectrics

- Capacitors and Capacitance (Continued)
- General Series and Parallel Connections

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

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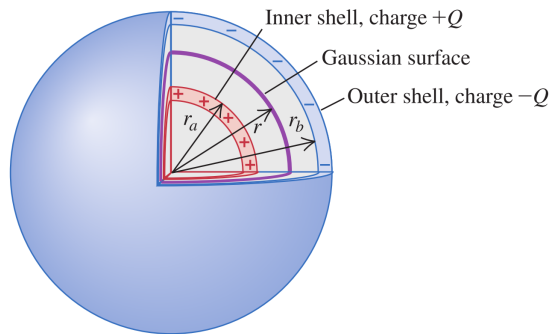
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Exercise I

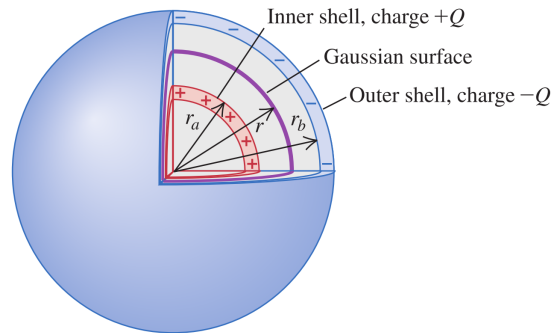
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Two concentric spherical conducting shells are separated by vacuum as shown in figure. The inner shell has total charge $+Q$ and outer radius r_a , and the outer shell has charge $-Q$ and inner radius r_b . Find the capacitance of this spherical capacitor.



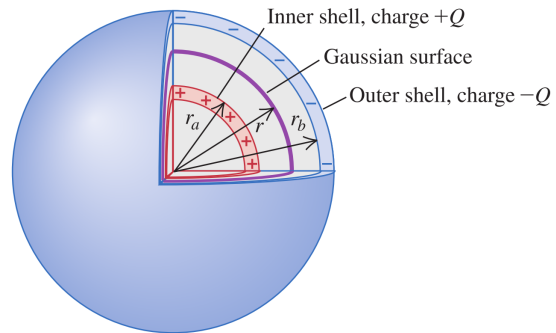
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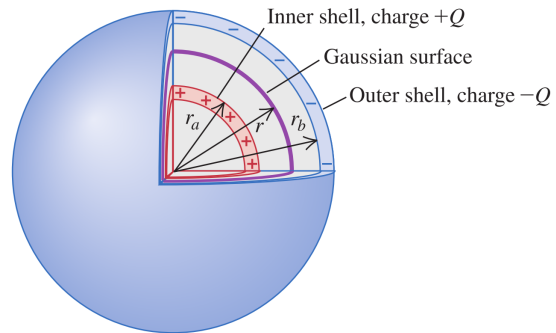
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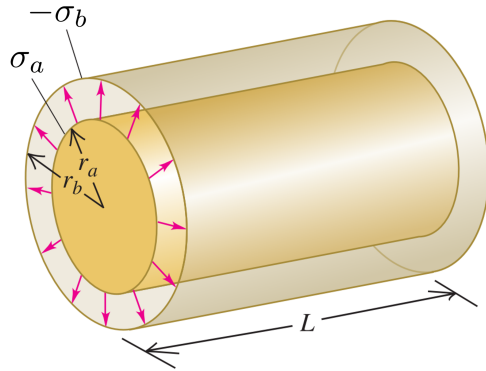
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Exercise 2

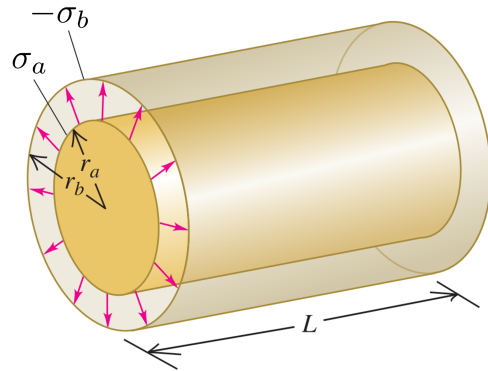
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Two long, coaxial cylindrical conductors are separated by vacuum as in figure. The inner cylinder has outer radius r_a and surface charge density σ_a . The outer cylinder has inner radius r_b and surface charge density $-\sigma_b$. Assume that the two surfaces have the same total charge Q in the figure. Find the capacitance per unit length for this capacitor.



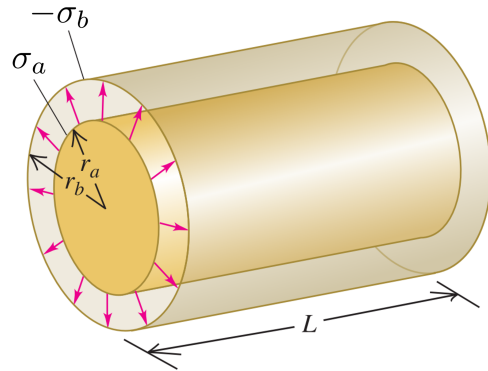
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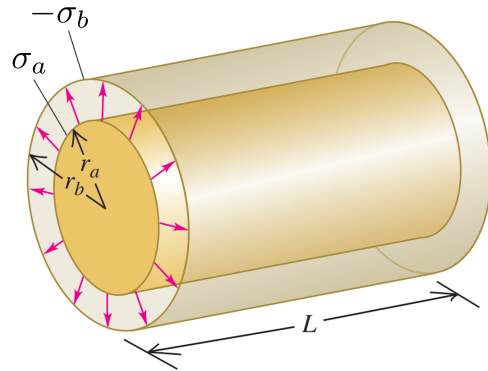
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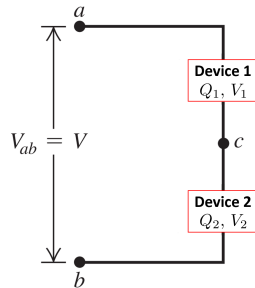
Series vs. Parallel Connection

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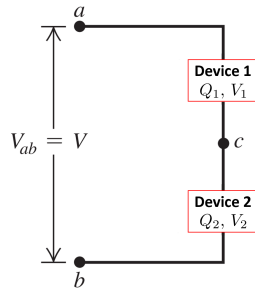
- **Series**: the same wire connects both components



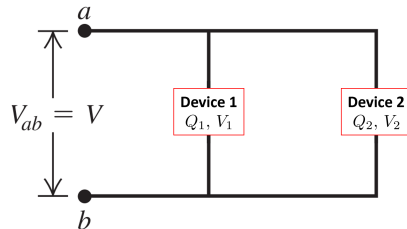
Series vs. Parallel Connection

Electronic components can be **connected** to each other in **two ways**:

- **Series**: the same wire connects both components



- **Parallel**: the wire from one component is branched into 2 or more wires before being connected to the other component



Series Connection

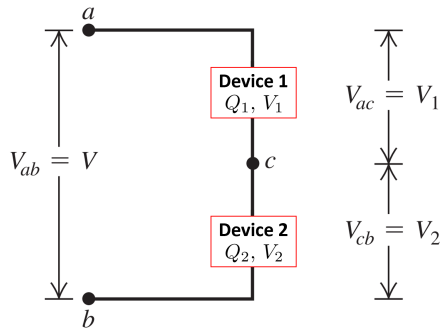
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Series Connection

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- **Charge (Q)** on, or **current (I)** passing through devices 1 and 2 **are the same**

$$Q_1 = Q_2 \Rightarrow I_1 = I_2$$



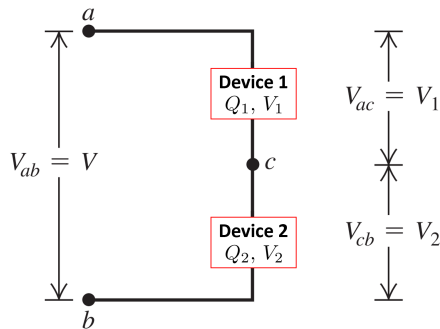
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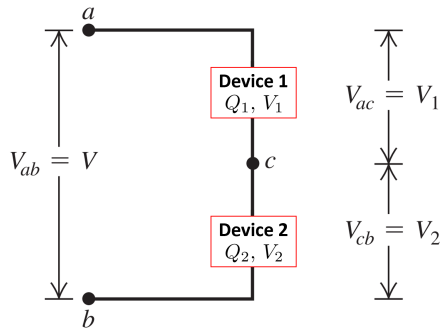
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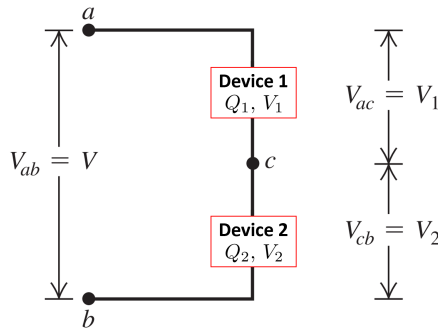
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- **Add up** together

$$\underbrace{V_b - V_a}_{V_{ab}} = \underbrace{V_c - V_a}_{V_{ac}=V_1} + \underbrace{V_b - V_c}_{V_{cb}=V_2} \Rightarrow V_{ab} = V_1 + V_2$$



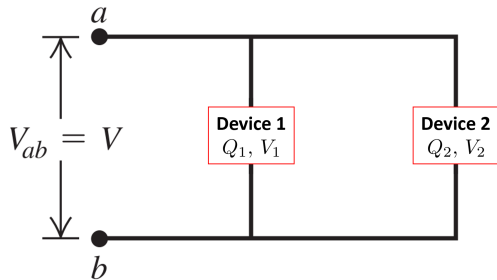
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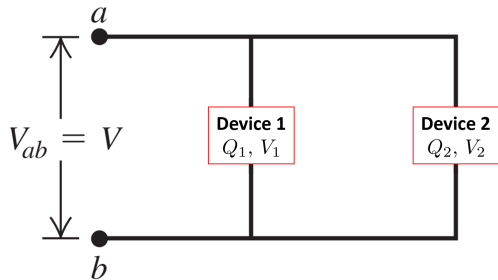


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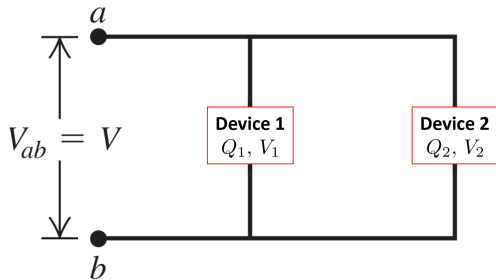
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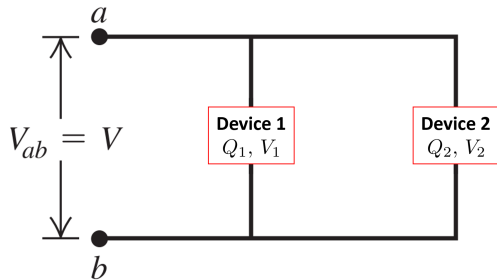
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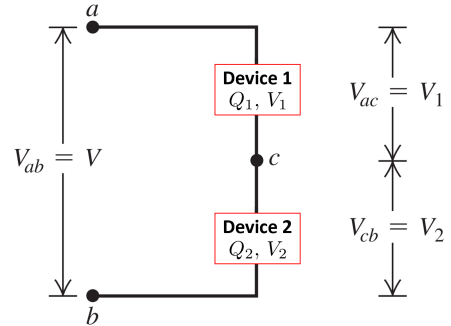
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Equivalence in Series Connection

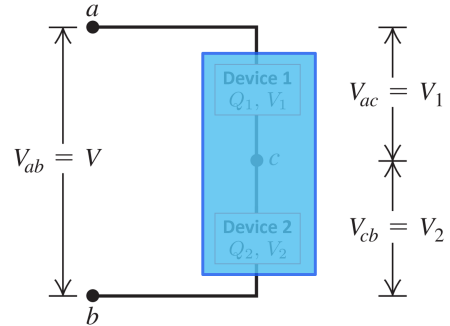
Equivalence in **series** connection:



Equivalence in Series Connection

Equivalence in **series** connection:

We can replace devices 1 and 2 with a single device



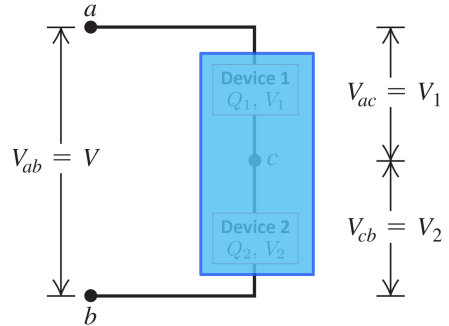
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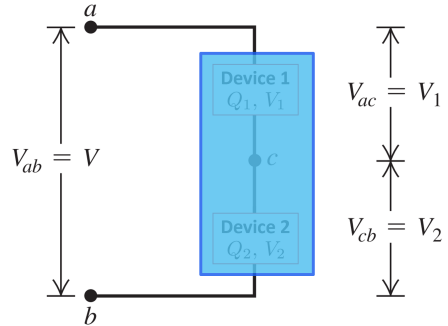
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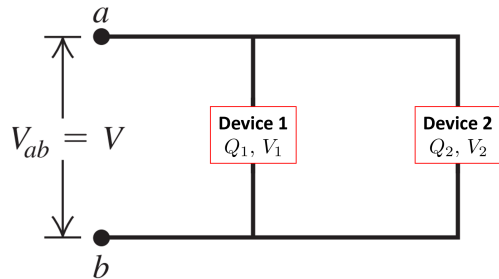
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$$I = I_1 = I_2$$



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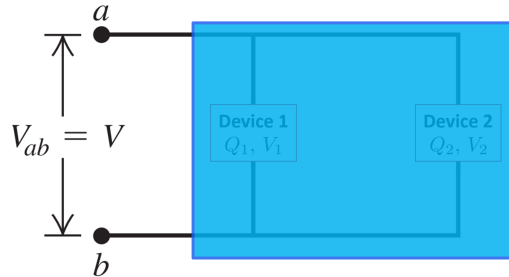
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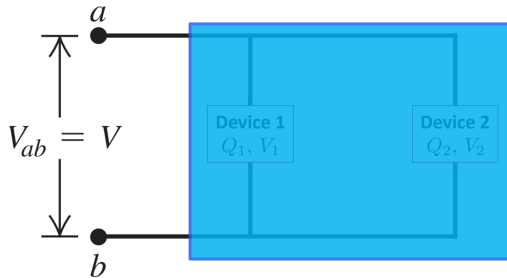
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